

1 Introduction

Viterbo’s conjecture states that among convex bodies of fixed volume, the ball has the largest symplectic capacity [Vit00]. We work in $\mathbb{R}^4$ with the Ekeland–Hofer–Zehnder capacity c_{EHZ} : for a convex body K , $c_{\text{EHZ}}(K)$ is the least action of a closed characteristic on ∂K , and on polytopes the minimum runs over generalized characteristics, which follow the facet normals. In dimension four the conjecture reads

$$\text{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)} \leq 1,$$

with equality for the ball. We call $\text{sys}(K)$ the systolic ratio. It is unchanged by translations, rescaling and linear symplectic maps.

Write $\mathbb{R}^4 = \mathbb{R}_q^2 \oplus \mathbb{R}_p^2$ with coordinates (q_1, q_2, p_1, p_2) . For planar convex bodies K_q, K_p , the Lagrangian product $K_q \times_L K_p$ is the set of points with $(q_1, q_2) \in K_q$ and $(p_1, p_2) \in K_p$. The relative position of the two factors affects capacity even though it does not affect volume. Haim–Kislev and Ostrover disproved the conjecture with a product of two regular pentagons [HO26]:

$$K_{\text{HKO}} = P_5 \times_L R(-\pi/2)P_5, \quad \text{sys}(K_{\text{HKO}}) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here P_5 is the regular pentagon of circumradius one centred at the origin with a vertex at $(1, 0)$, and $R(\theta)$ is planar rotation by θ .

This thesis picks a toolbox and asks how much progress it makes on questions around Viterbo’s conjecture after this counterexample. The toolbox consists of high-performance computation on the LICCA cluster, proof by exact computation, standard data-science methods and AI agents. The agents are part of the method: they ran the data science and did large parts of the theory work. Pure mathematics has used data before: in knot theory, patterns spotted in computed invariants have suggested results that were then proved. Here AI both throws data science at the problem and develops much of the theory. The questions came out of the work, and the results answer different ones.

One of the earliest questions was whether standard data science finds new interesting bodies with systolic ratio above one. It did not. Bodies with systolic ratio above one appeared as small perturbations of K_{HKO} , none with a larger ratio; random samples and local searches from generic starting bodies all stayed below one. The data did show a pattern: in sampled populations, symplectic ridge area, the total symplectic area of the two-dimensional faces normalized by volume, is strongly inversely associated with the systolic ratio, although the association weakens among bodies selected for low ridge area or high systolic ratio.

Because random perturbations did not improve K_{HKO} , we conjectured that it is a local maximum, and we prove this among polytopes with ten facets: every sufficiently close ten-facet polytope has systolic ratio at most $\text{sys}(K_{\text{HKO}})$, with equality only for translations, rescalings and linear symplectic images. The same ten-facet space contains a second, non-strict local maximum at systolic ratio one, a Lagrangian product of a hexagon and a quadrilateral. Within Lagrangian products of two independently linearly deformed regular pentagons, we prove an explicit capacity formula and the sharp bound $(3 + \sqrt{5})/5$. These proofs rest on Haim–Kislev’s finite formula for the capacity of a polytope [Hai19], its first-order variation in the facet data, and exact arithmetic.

Section 1.1 places the EHZ capacity and the counterexample among related results on capacities, dynamics and convex geometry. Sections 1.2–1.4 describe the local theorem, the finite capacity computations and the data-science investigation in more detail, and Section 1.5 explains how to read the thesis.

References

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- [Vit00] Claude Viterbo. “Metric and isoperimetric problems in symplectic geometry”. In: *Journal of the American Mathematical Society* 13.2 (2000), pp. 411–431. DOI: 10.1090/S0894-0347-00-00328-3.